

# Aakash KT

<https://aakashkt.github.io/>

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## Experience

Dec 2025 - **Senior Graphics Research Engineer**, *Huawei IRC*, Netherlands.

Present Part of the Game Rendering Lab, with projects on identifying and resolving bottlenecks in real-time mobile games, using novel rendering algorithms and hardware specific optimizations, under power & performance constraints of mobile devices.

Jun 2024 - **Senior Graphics Research Engineer**, *Qualcomm*, India.

Oct 2025 Pre-silicon research explorations for Qualcomm's next-generation Adreno GPUs and GPU acceleration structures research.  
Patent filed on improving BVH traversal performance in SIMD architectures.

Aug 2022 - **Research Scientist Intern**, *Meta Reality Labs*, Pittsburgh, USA.

Jan 2023 **Manager:** Dr. Giljoo Nam

Project on accelerating photorealistic rendering of human hair with multiple scattering, resulting in a manuscript publication at EGSR 2023 with published OptiX & CUDA code.

2017 - 2020 **Research Assistant**, *CVIT, IIIT Hyderabad*, India.

**Advisor:** Dr. P. J. Narayanan

Neural Rendering for specific usecases and appearance editing from captured photographs. Additional projects on depth estimation from focus images and explored deep generative models for domain adaptation.

## Education

2020 - 2024 **Ph.D in Computer Science**, *IIIT Hyderabad*.

**Advisor:** Prof. Dr. P. J. Narayanan, **CGPA:** 8.0/10.0

Thesis on accelerating physically based rendering via analytic solutions and efficient neural approximations of the rendering equation.

2015 - 2020 **BTech and MS by Research in Computer Science**, *IIIT Hyderabad*.

**Advisor:** Prof. Dr. P. J. Narayanan, **CGPA:** 7.21/10.0

**Relevant courses:** Computer Graphics, Computer Vision, Digital Image Processing, Artificial Intelligence, Statistical Methods in AI, Optimization Methods, Advanced Computer Networks, Operating Systems, Software Engineering.

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## Selected Publications

- Eurographics 2025 **Linearly Transformed Spherical Distributions for Interactive Single Scattering with Area Lights**, *CGF, Full Paper*.  
**Aakash KT**, Ishaan Shah, P. J. Narayanan
- SIGGRAPH Asia 2023 **Combining Resampled Importance & Projected Solid Angle Samplings for Many Area Light Rendering**, *Tech. Comm.*  
Ishaan Shah\*, **Aakash KT\***, P. J. Narayanan
- EGSR 2023 **Accelerating Hair Rendering by Learning High-Order Scattered Radiance**, *CGF, Full Paper*.  
**Aakash KT**, Adrian Jarabo, Carlos Aliaga, Matt Jen-Yuan Chiang, Olivier Maury, Christophe Hery, P. J. Narayanan, Giljoo Nam
- I3D 2022 **Bringing Linearly Transformed Cosines to Anisotropic GGX**, *Best Paper Award*.  
**Aakash KT**, Eric Heitz, Jonathan Dupuy, P. J. Narayanan
- EGSR 2021 **Fast Analytic Soft Shadows from Area Lights**, *Full Paper*.  
**Aakash KT**, Parikshit Sakurikar, P. J. Narayanan
- SIGGRAPH Asia 2019 **A Flexible Neural Renderer for Material Visualization**, *Technical Brief*.  
**Aakash KT**, Parikshit Sakurikar, Saurabh Saini, P. J. Narayanan

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## Achievements & Activities

- 2022 **Best Paper Award**, *I3D 2022*.  
Received NVIDIA RTX 3090 as the prize.
- 2022 **Tertiary Reviewer**, *ICVGIP 2022*.
- 2021 **Tertiary Reviewer**, *Pacific Graphics 2021*.
- 2020 **KCIS Ph.D fellowship**.  
Received the KCIS Ph.D fellowship for my research.
- 2019 **Microsoft Research Travel Grant**.  
Received a travel grant from Microsoft to present my work at **SIGGRAPH Asia 2019**.
- 2018 **CANSAT competition, NASA, Texas, USA**.  
Participated in the CANSAT competition in which teams build a payload that is released from a height of thousand meters. The task is to perform various maneuvers and get back to land, all without damaging an egg kept inside the payload. **Secured a world rank of 24 as co-team leader**.